2020/2021 RPM Base Residual Auction Results

Executive Summary

The 2020/2021 Reliability Pricing Model (RPM) Base Residual Auction (BRA) cleared 165,109.2 MW of unforced capacity in the RTO representing a 23.9% reserve margin. Accounting for load and resource commitments under the Fixed Resource Requirement (FRR), the reserve margin for the entire RTO for the 2020/2021 Delivery Year as procured in the BRA is 23.3%, or 6.7% higher than the target reserve margin of 16.6%. This reserve margin was achieved at clearing prices that are between approximately 26% to 66% of Net CONE, depending upon the Locational Deliverability Area (LDA), while attracting 2,350 MW of new combined cycle gas resources.

The 2020/2021 BRA is the first where PJM has procured 100% Capacity Performance ("CP") Resources. CP Resources must be capable of sustained, predictable operation, and are expected to be available and capable of providing energy and reserves when needed throughout the entire Delivery Year. Also, the 2020/2021 BRA was conducted under the provisions of PJM's Enhanced Aggregation filing (Docket ER17-367-000 & 001) which was accepted by FERC on March 21, 2017.

2020/2021 BRA Resource Clearing Prices

Resource Clearing Prices (RCPs) for the 2020/2021 BRA are shown in Table 1 below. The RCP for CP Resources located in the rest of RTO is $76.53/MW-day. The MAAC LDA, EMAAC LDA, ComEd LDA and DEOK LDA were constrained LDAs in the 2020/2021 BRA with locational price adders of $9.51/MW-day, $101.83/MW-day, $111.59/MW-day and $53.47/MW-day, respectively, for all resources located in those LDAs. For comparison purposes, the RCP for CP Resources located in the rest of RTO and MAAC in the 2019/2020 BRA was $100.00/MW-day. For the same year, the RCP for CP Resources in the EMAAC LDA was $119.77/MW-day and the RCP for CP Resources in the COMED LDA was $202.77 /MW-day in the 2019/2020 BRA. The DEOK LDA was not modeled in the 2019/20 BRA and cleared with the rest of RTO.

2020/2021 RPM Base Residual Auction Results

2020/2021 BRA Cleared Capacity Resources

As seen in the table below, the 2020/2021 BRA procured 2,389.3 MW of capacity from new generation and 434.5 MW from uprates to existing or planned generation. The quantity of capacity procured from external Generation Capacity Resources in the 2020/2021 BRA is 3,997.2 MW which is an increase of 121.3 MW from that procured in last year's BRA. All external generation capacity that has cleared in the 2020/21 BRA has the requirements for the Capacity Import Limit (CIL) exception which include (1) long-term firm transmission service has been confirmed on the complete transmission path from the external resource into PJM for the relevant Delivery Year; (2) the external resource meets or will meet prior to the Delivery Year all applicable requirements to be pseudo-tied; and (3) a separate written commitment has been executed to offer all unforced capacity of the external resource into RPM Auctions under the same terms, and subject to the same conditions and exceptions, as set forth for internal generation resources by section 6.6 of Attachment DD of PJM Tariff. The total quantity of DR procured in the 2020/2021 BRA is 7,820.4 MW which is a decrease of 2,527.6 MW from that procured in last year's BRA; and, the total quantity of EE procured in the 2020/2021 BRA is 1,710.2 MW, which is an increase of 195.1 MW from that procured in last year's BRA.

Megawatts of Unforced Capacity Procured by Type from the 2014/2015 BRA to the 2020/2021 BRA

2020/2021 RPM Base Residual Auction Results

Introduction

This document provides information for PJM stakeholders regarding the results of the 2020/2021 Reliability Pricing Model (RPM) Base Residual Auction (BRA). The 2020/2021 BRA opened on May 10, 2017, and the results were posted on May 23, 2017.

In each BRA, PJM seeks to procure a target capacity reserve level for the RTO in a least cost manner while recognizing the following reliability-based constraints on the location and type of capacity that can be committed:

- Internal PJM locational constraints are established by setting up Locational Deliverability Areas (LDAs) with each LDA having a separate target capacity reserve level and a maximum limit on the amount of capacity that it can import from resources located outside of the LDA.
- Total cleared summer-period sell offers must exactly equal total cleared winter-period sell offers across the entire RTO to ensure that seasonal CP sell offers clear to form annual CP commitments.

The auction clearing process commits capacity resources to procure a target capacity reserve level for the RTO in a least-cost manner while recognizing and enforcing these reliability-based constraints. The clearing solution may be required to commit capacity resources out-of-merit order but again in a least-cost manner to ensure that all of these constraints are respected. In those cases where one or more of the constraints results in out-of-merit commitment in the auction solution, resource clearing prices will be reflective of the price of resources selected out of merit order to meet the necessary requirements.

This document begins with a high-level summary of the BRA results followed by sections containing detailed descriptions of the 2020/2021 BRA results and a discussion of the results in the context of the ten previous BRAs.

Summary of Results

The 2020/2021 Reliability Pricing Model (RPM) Base Residual Auction (BRA) cleared 165,109.2 MW of unforced capacity in the RTO representing a 23.9% reserve margin. The reserve margin for the entire RTO is 23.3%, or 6.7% higher than the target reserve margin of 16.6%, when the Fixed Resource Requirement (FRR) load and resources are considered.

Resource Clearing Prices (RCPs) for the 2020/2021 BRA are shown in Table 1 below. The RCP for CP Resources located in the rest of RTO is $76.53/MW-day. The MAAC LDA, EMAAC LDA, ComEd LDA and DEOK LDA were constrained LDAs in the 2020/2021 BRA with locational price adders of $9.51/MW-day, $101.83/MW-day, $111.59/MW-day and $53.47/MW-day, respectively, for all resources located in those LDAs. For comparison purposes, the RCP for CP Resources located in the rest of RTO